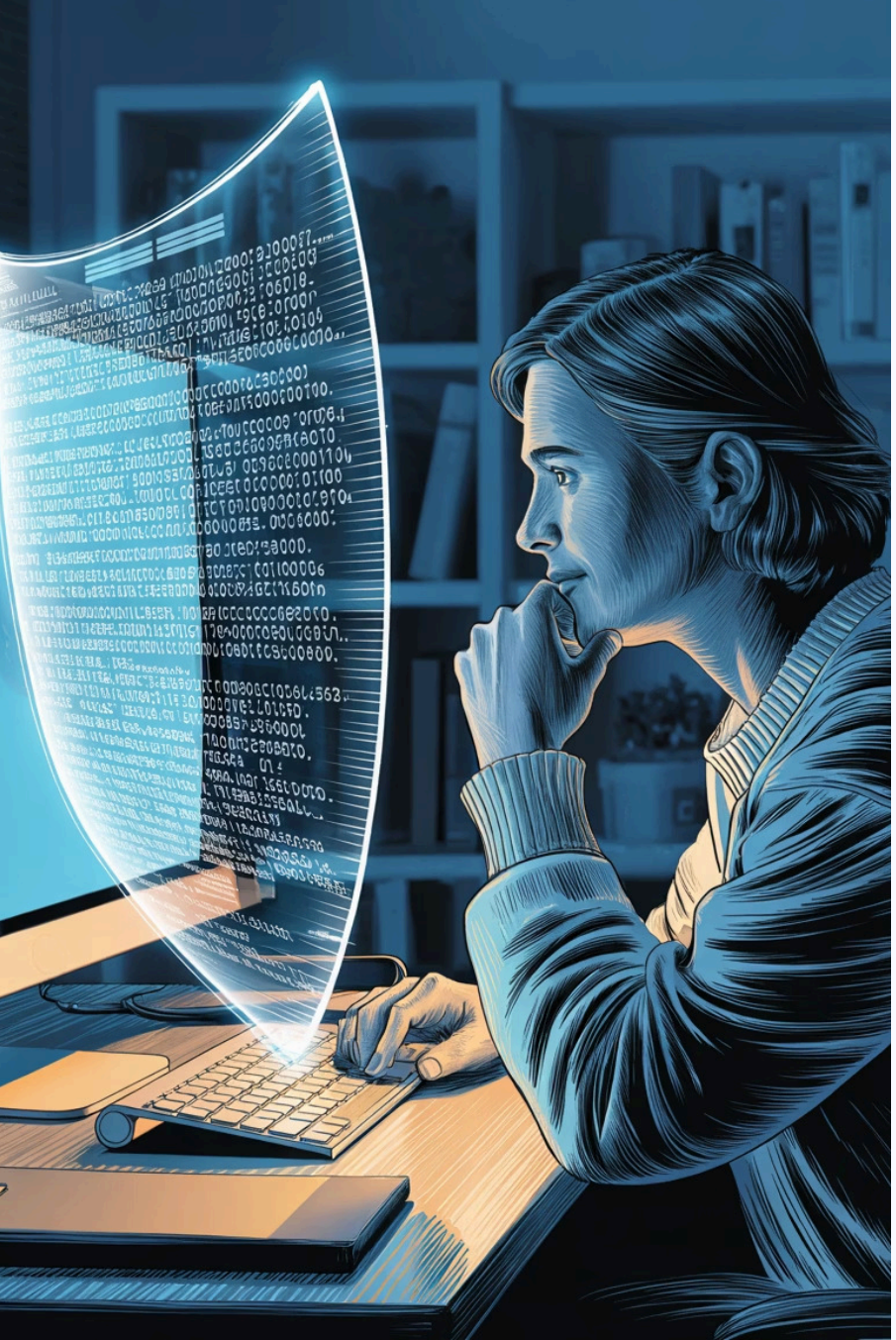

Privacy Guard

A Google Chrome browser extension designed to protect users' personal data while browsing. It provides a comprehensive suite of tools to block trackers, analyze privacy policies, and prevent browser fingerprinting.

Team Members

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Core Features

The project offers 6 main features to ensure user privacy:

Tracker Detection & Blocking

Cookie Management

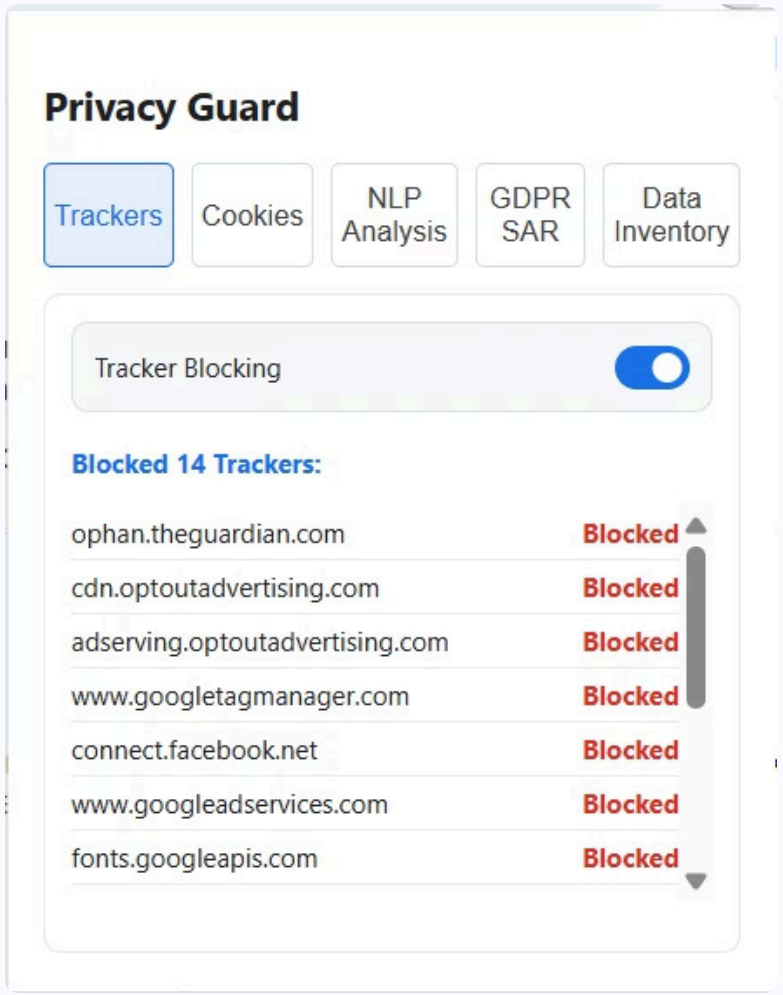
Fingerprinting Prevention

NLP Privacy Policy Analysis

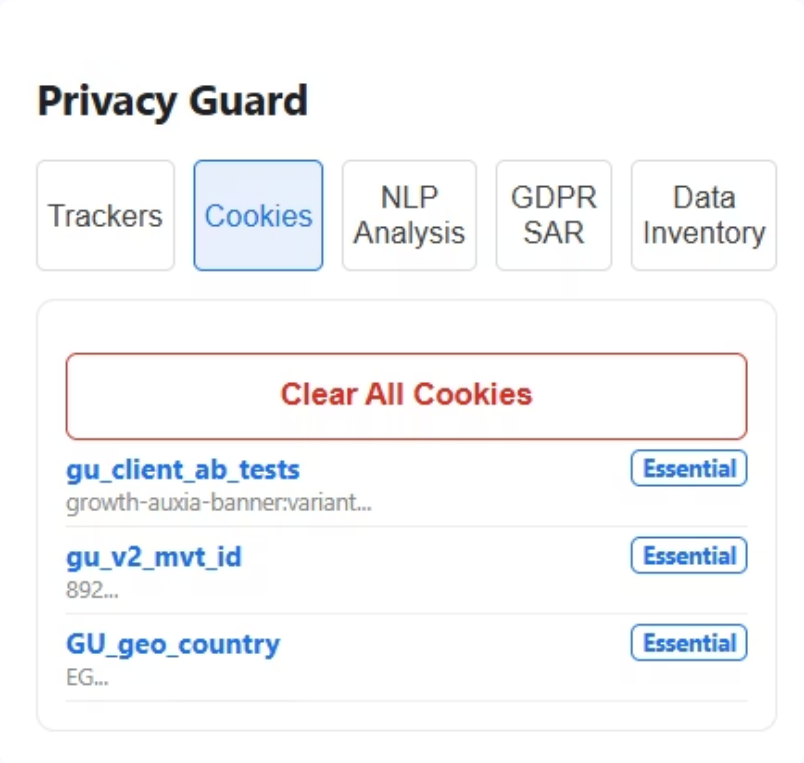
Personal Data Inventory

GDPR Legal Request Templates

Tracker Detection & Blocking



- Operates automatically in the background on any visited website
- Intercepts outgoing requests and compares them against the built-in easyprivacy.txt list
- Instantly blocks requests from known tracking domains and logs them for the user to see
- Can be toggled on or off easily from the extension popup



Cookie Management

View All Cookies

Displays all cookies currently active on the visited website

Smart Classification

Automatically classifies cookies as "Tracking" (if the name contains _ga, ads, track, etc.) or "Essential"

One-Click Clear

Features a one-click "Clear All Cookies" button to instantly website data

Fingerprinting Prevention

Protects against 4 different methods websites use to uniquely identify your device:

Canvas Fingerprinting

Modifies pixel data to prevent unique visual identification

WebGL Fingerprinting

Returns generic vendor data instead of your actual graphics card details

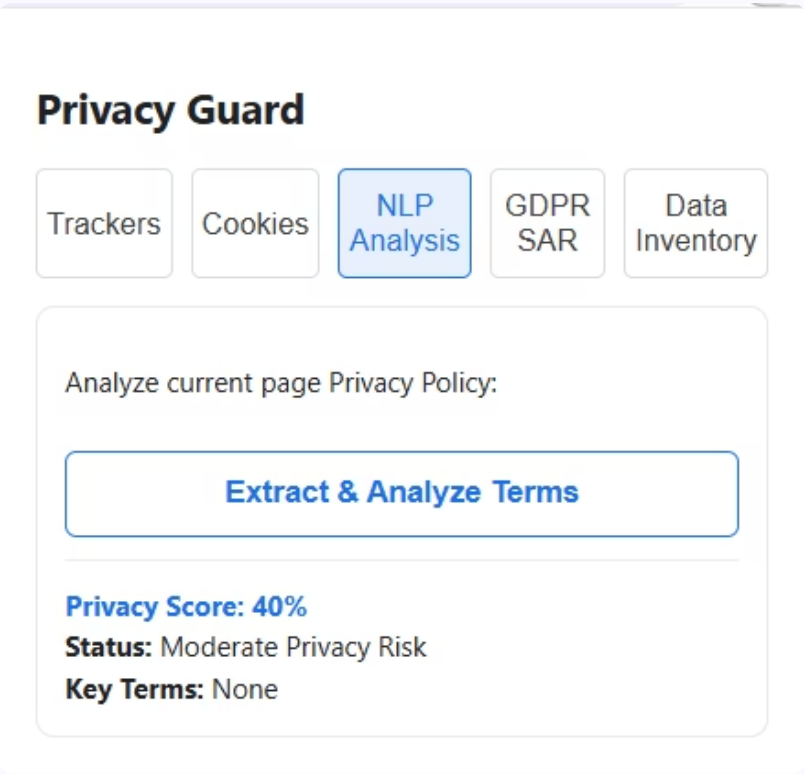
AudioContext Fingerprinting

Injects minor random noise into audio frequency data

Navigator Properties

Spoofs hardware information like OS, memory, and CPU cores

NLP Privacy Policy Analysis

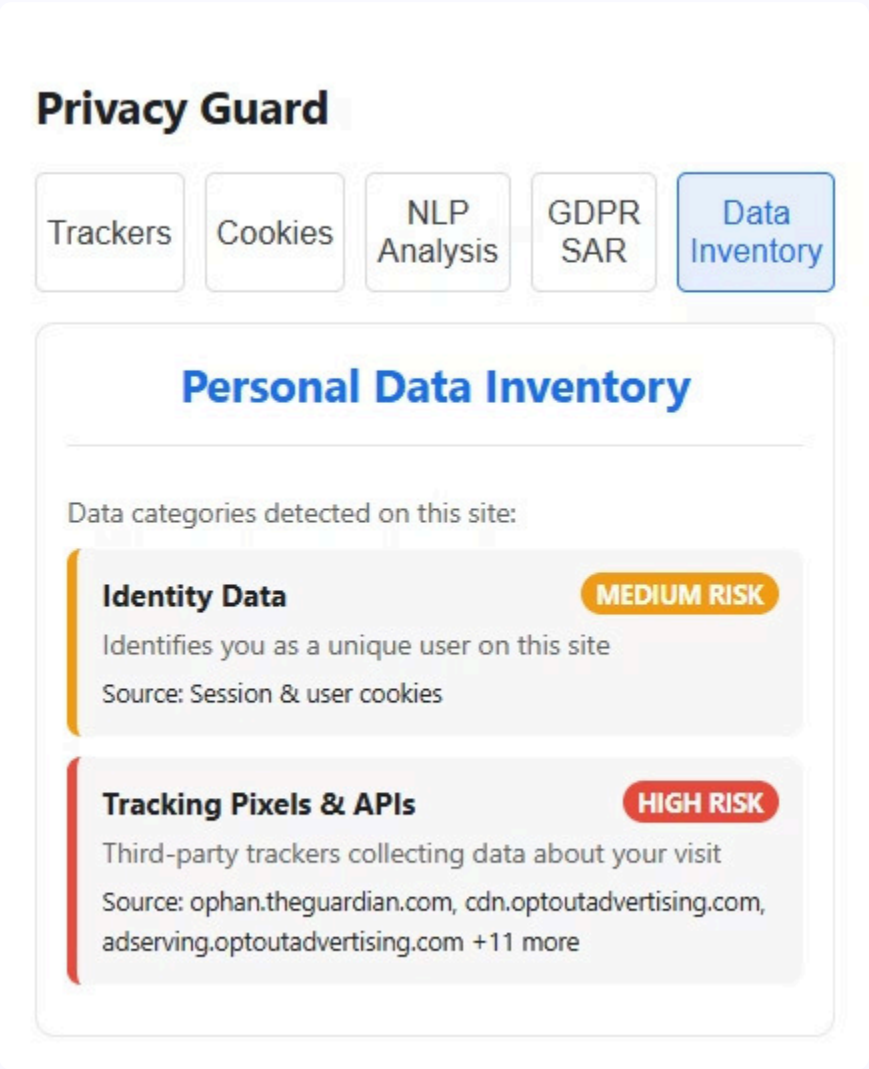


How It Works

1. **Scans website "Privacy Policy" pages to extract the full text**
2. **Sends the text to a Python (Flask) backend server**
3. **Uses Natural Language Processing (NLTK) to analyze the text and generate a Privacy Score (0 to 100)**

The score increases with privacy-positive keywords (e.g., security, consent) and decreases with privacy-negative keywords (e.g., tracking, third-party, share).

Personal Data Inventory



Scans trackers and cookies to build a categorized list of the personal data the website is collecting.

Low Risk

Preferences and language data

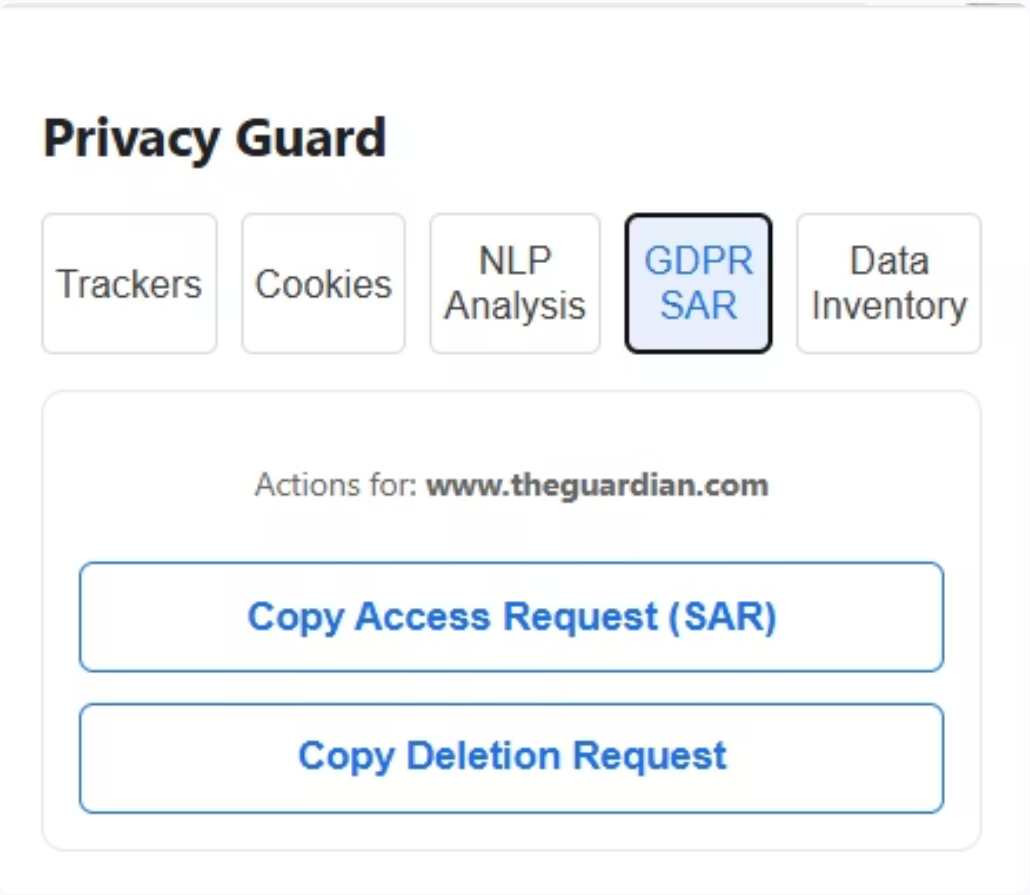
Medium Risk

Identity and session data

High Risk

Behavioral data and tracking pixels

GDPR Legal Request Templates



Provides ready-to-use legal email templates based on European GDPR laws. Automatically fills in the current website's domain name.

Subject Access Request (SAR)

Deletion Request (Right to be Forgotten)

Technology Stack

Frontend

HTML5, CSS3, JavaScript, WebExtensions API (Manifest V3)

Storage

IndexedDB (for local, secure storage of tracker data)

Backend

Python, Flask, Flask-CORS

AI / NLP

NLTK (Natural Language Toolkit) for text tokenization and scoring

Data Source

EasyPrivacy blocklist

Thank you

Any questions?